

SOP Purely NAND Seven Segment

ACTIVITY 11

ALEGRE, CHRIS
22-3452
BSIT 4-6

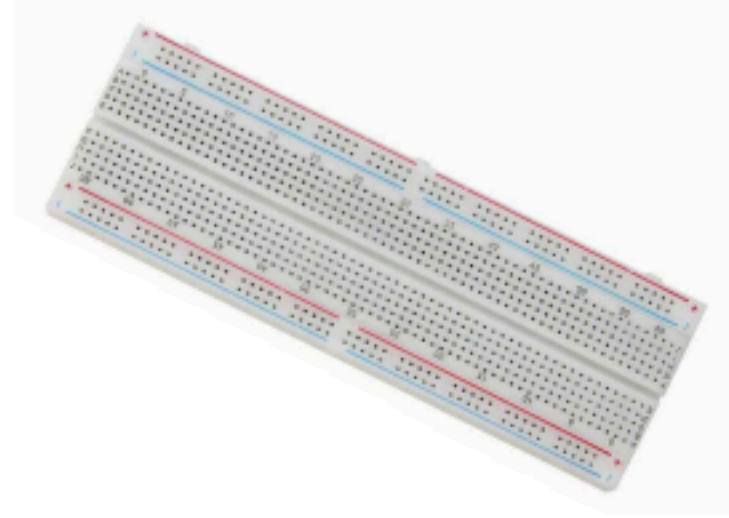


OBJECTIVE

- To design, simulate, and implement a combinational logic circuit that displays a specific sequence of 8 alphanumeric characters (`a,l,e,c,h,r,L,C`) on a standard 7-segment display module.
- To restrict the logic circuit design exclusively to Universal NOR gates for all required boolean inversions, conjunctions, and disjunctions.
- To derive and optimize 7 independent boolean expressions representing the display segments (\$a, b, c, d, e, f, g\$) using Boolean algebra, Karnaugh Maps (K-Maps), and Product-of-Sums (POS) criteria to natively suit NOR-gate architecture.
- To verify the functional accuracy of the NOR-only gate network by successfully cycling through the inputs and cleanly illuminating the sequence on a physical or simulated common-cathode/common-anode display.

MATERIALS NEEDED

Breadboard



Used as a temporary platform for assembling and testing the logic circuit without soldering, allowing easy modification and troubleshooting of connections.

Jumper Wire (Male to Male)



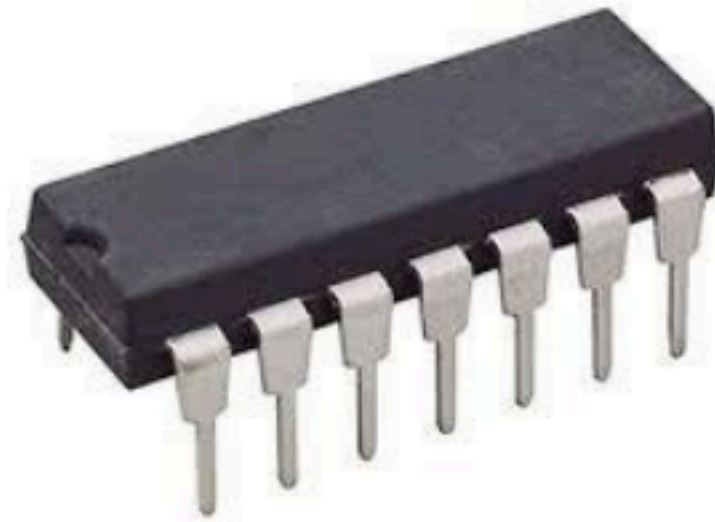
Used to interconnect the ICs, LEDs, and power supply on the breadboard to establish proper electrical connections.

Power Supply



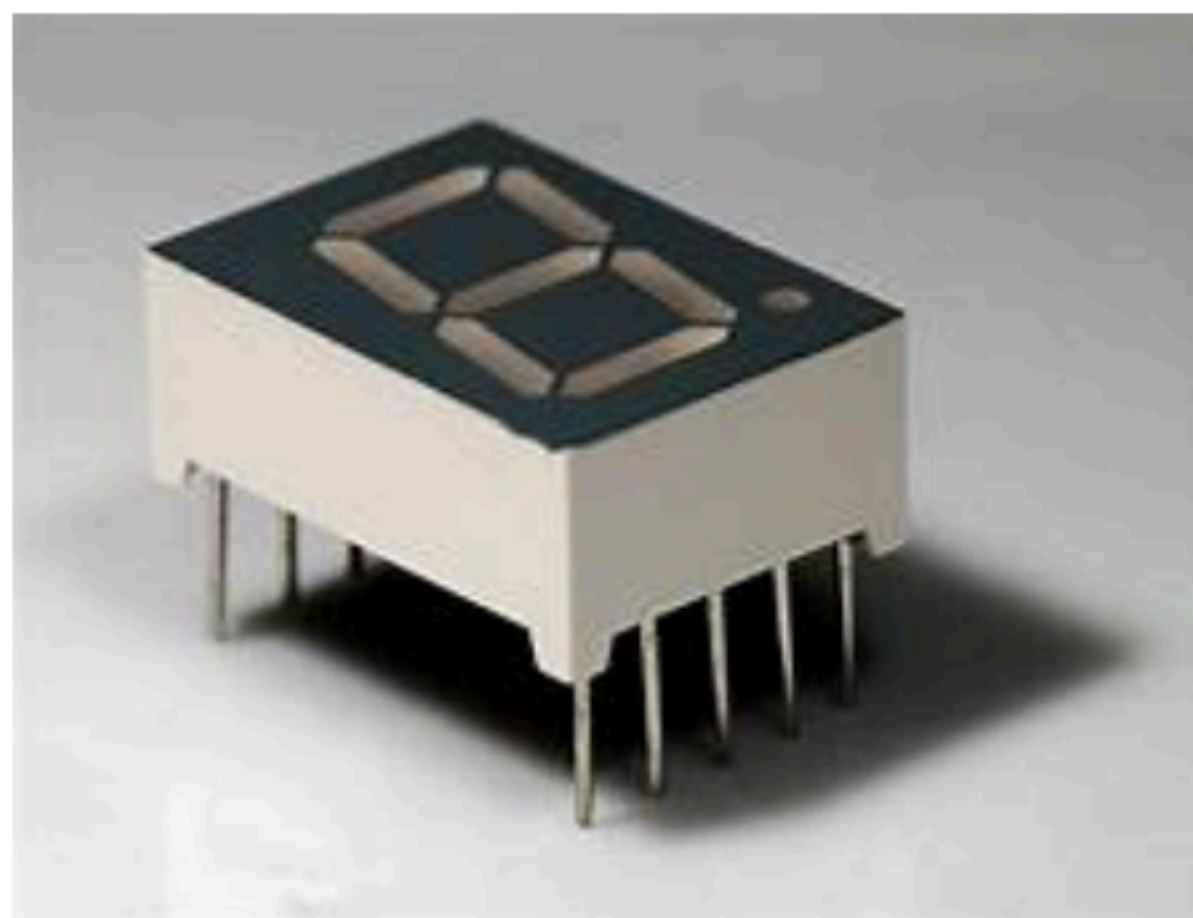
Provides the required operating voltage (typically +5V for TTL ICs) necessary for proper functioning of the logic integrated circuits.

Integrated Circuits (ICs):



74LS32 (**OR** Gate)

Contains multiple NAND gates used to perform NAND logic operations within the circuit.



A 7-segment display is an electronic display device used to show numerical digits (0–9) using seven individual segments arranged in a rectangular pattern. Each segment is typically an LED that can be turned on or off to form different numbers.

PROCEDURES

Step 1: Character and Segment Mapping Analyze the target character sequence (`a,l,e,c,h,r,L,C`) and determine the required active segments (\$a\$ through \$g\$) for each character on a standard 7-segment display, carefully selecting uppercase or lowercase shapes to ensure maximum legibility.

Step 2: Truth Table Formulation Construct a comprehensive truth table mapping the 3-bit binary inputs (X, Y, Z) across eight states to the seven output segments, marking a 1 for an active segment and a 0 for an inactive segment.

Step 3: Boolean Simplification via POS Derive the Product-of-Sums (POS) expressions for each segment by identifying the Maxterms (where the output is 0). Use Karnaugh Maps (K-Maps) to simplify the expressions into their minimal algebraic forms.

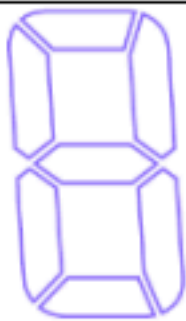
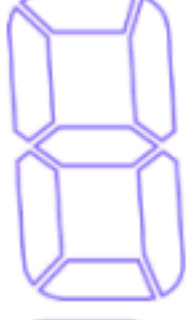
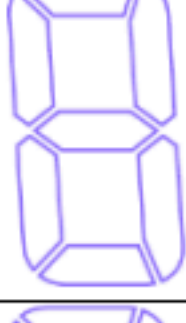
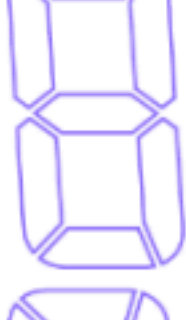
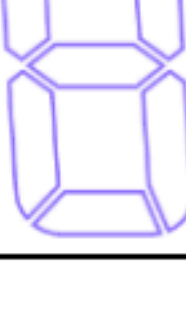
Step 4: NOR Logic Conversion Apply De Morgan's Laws and double inversion ($\overline{\overline{A}}$) to the simplified POS expressions, converting all standard AND/OR logic operations exclusively into NOR gate equivalents.

Step 5: Circuit Schematic Simulation Draft the circuit schematic in a digital logic simulator, configuring NOR gates as NOT gates (by tying inputs together) to generate inverted inputs, and wiring the combinational network for all seven segment outputs.

Step 6: Hardware Breadboard Assembly Translate the simulated design to a physical breadboard by placing the required NOR gate Integrated Circuits (ICs), routing the 3-bit input switches, and connecting the logic outputs to a common-cathode 7-segment display through current-limiting resistors.

Step 7: Testing and Troubleshooting Apply power to the circuit and systematically cycle through the binary inputs to verify that the display correctly and cleanly outputs the intended characters in the exact sequence.

Chris Castillejos Alegre

Cell: No.	A	B	C	DISPLAY	Appearance on 7 segments	A	B	C	D	E	F	G
0	0	0	0	a		1	1	1	1	1	0	1
1	0	0	1	l		0	1	1	0	0	0	0
2	0	1	0	e		1	1	0	1	1	1	1
3	0	1	1	c		0	0	0	1	1	0	1
4	1	0	0	h		0	0	1	0	1	1	1
5	1	0	1	r		0	0	0	0	1	0	1
6	1	1	0	L		0	0	0	1	1	1	0
7	1	1	1	C		1	0	0	1	1	1	0

For A

SoP (1s)

AB

00	01	11	10
0	2	6	4
1	3	7	5

Select answer #1 / 1

$S = ABC + \bar{A}\bar{C}$

For B

SoP (1s)

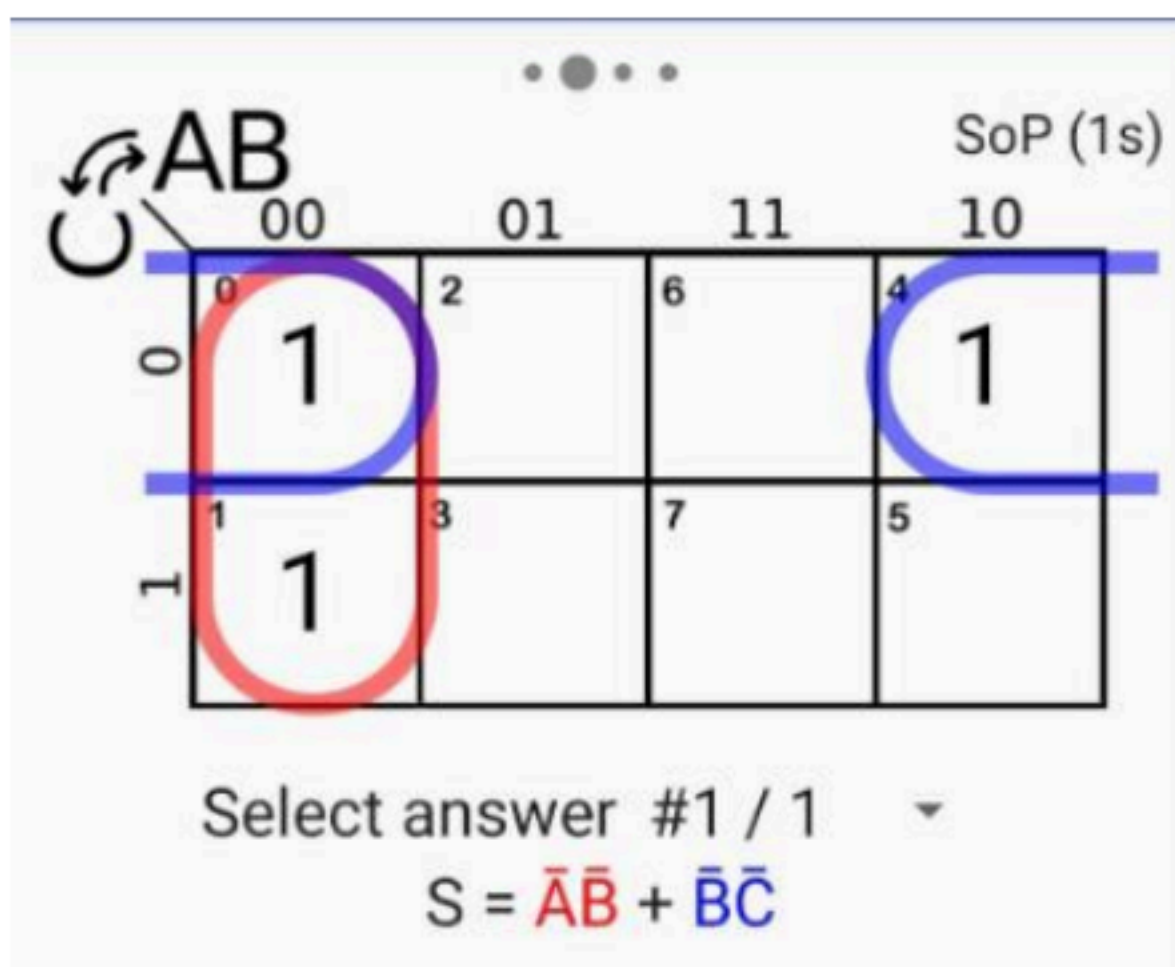
AB

00	01	11	10
0	2	6	4
1	3	7	5

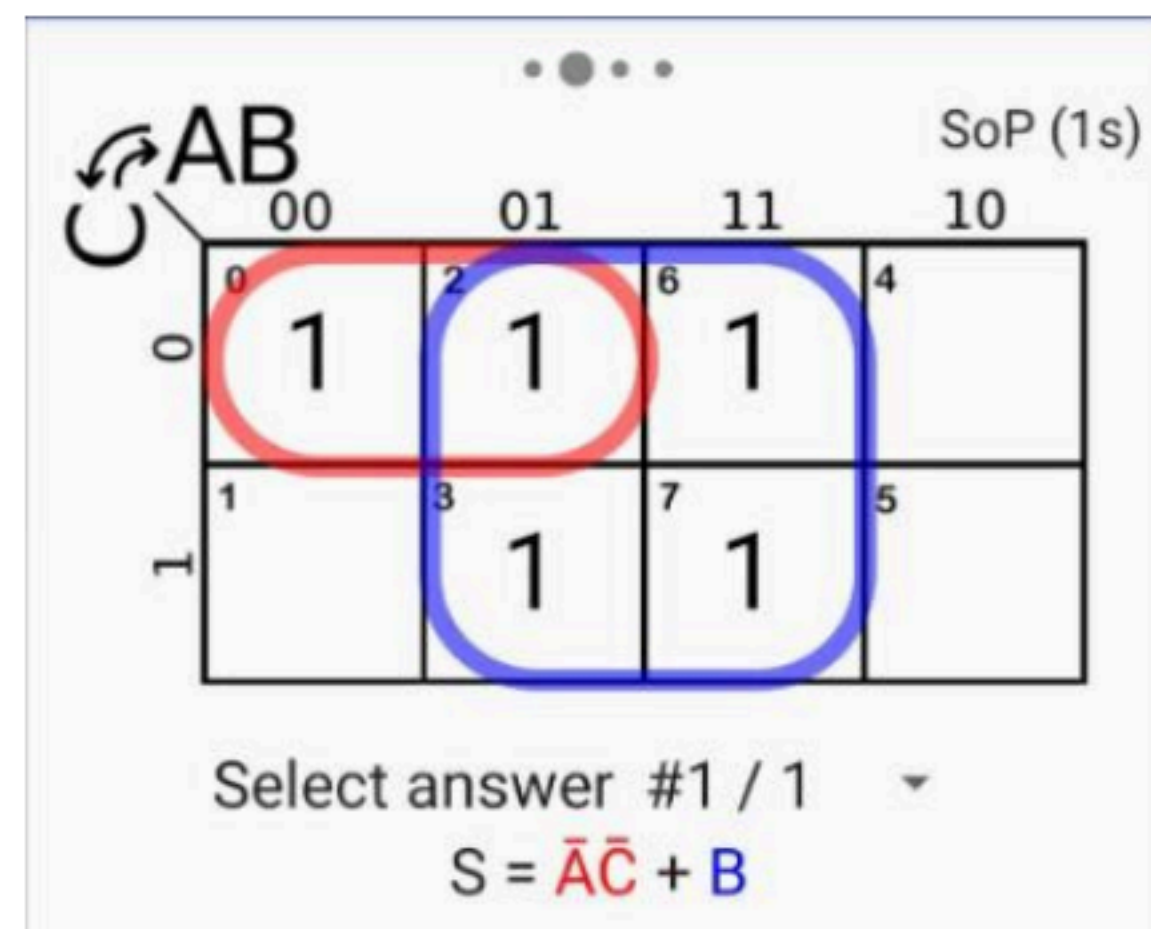
Select answer #1 / 1

$S = \bar{A}\bar{B} + \bar{A}\bar{C}$

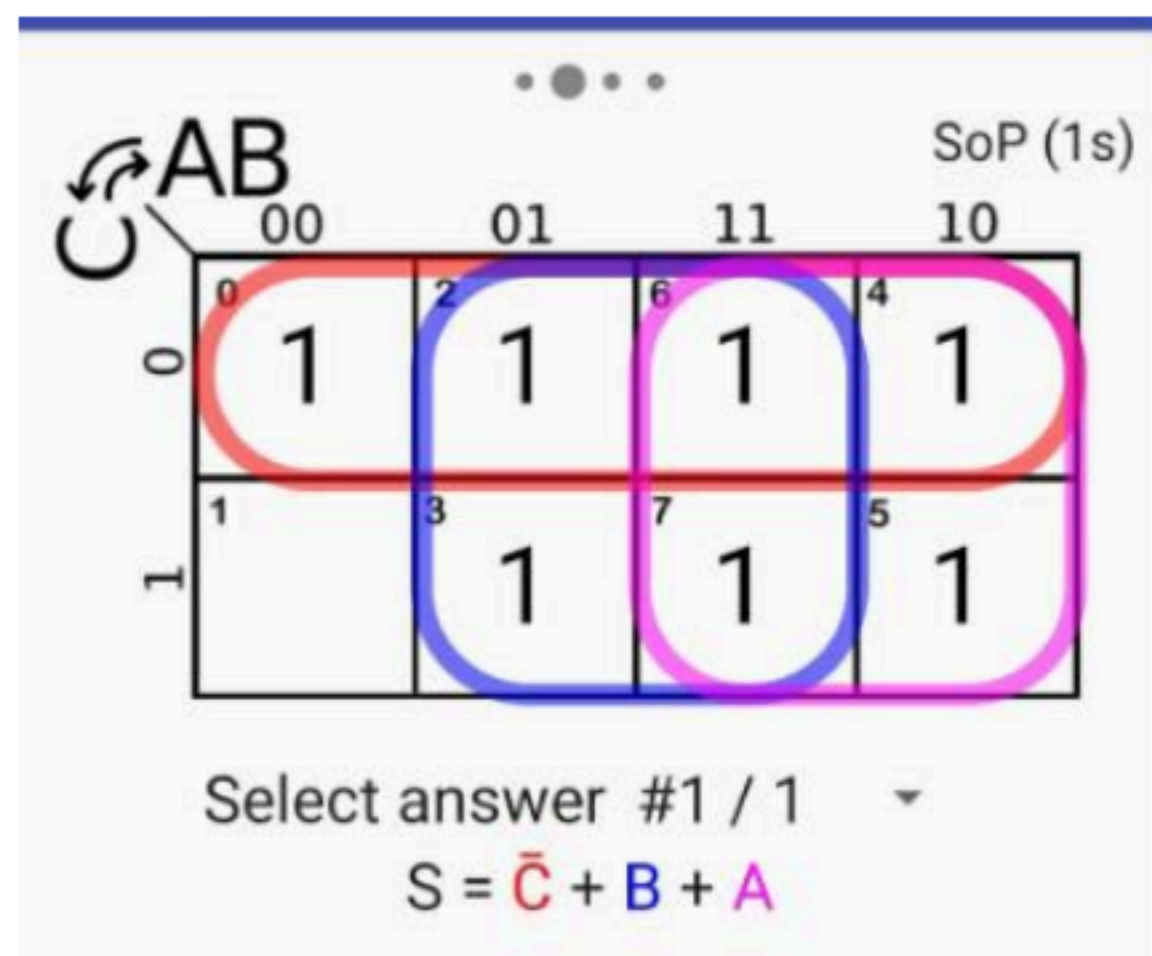
For C



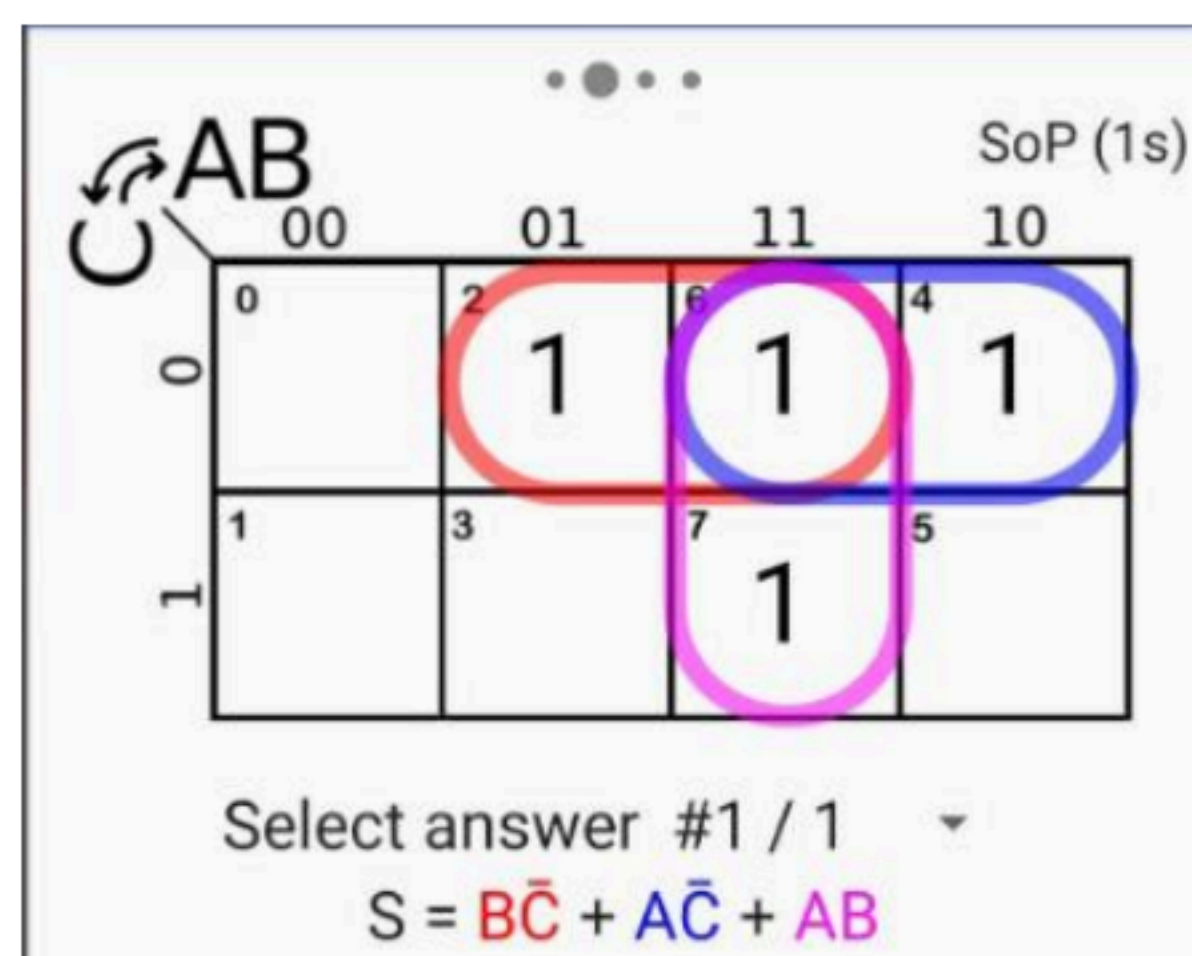
For D



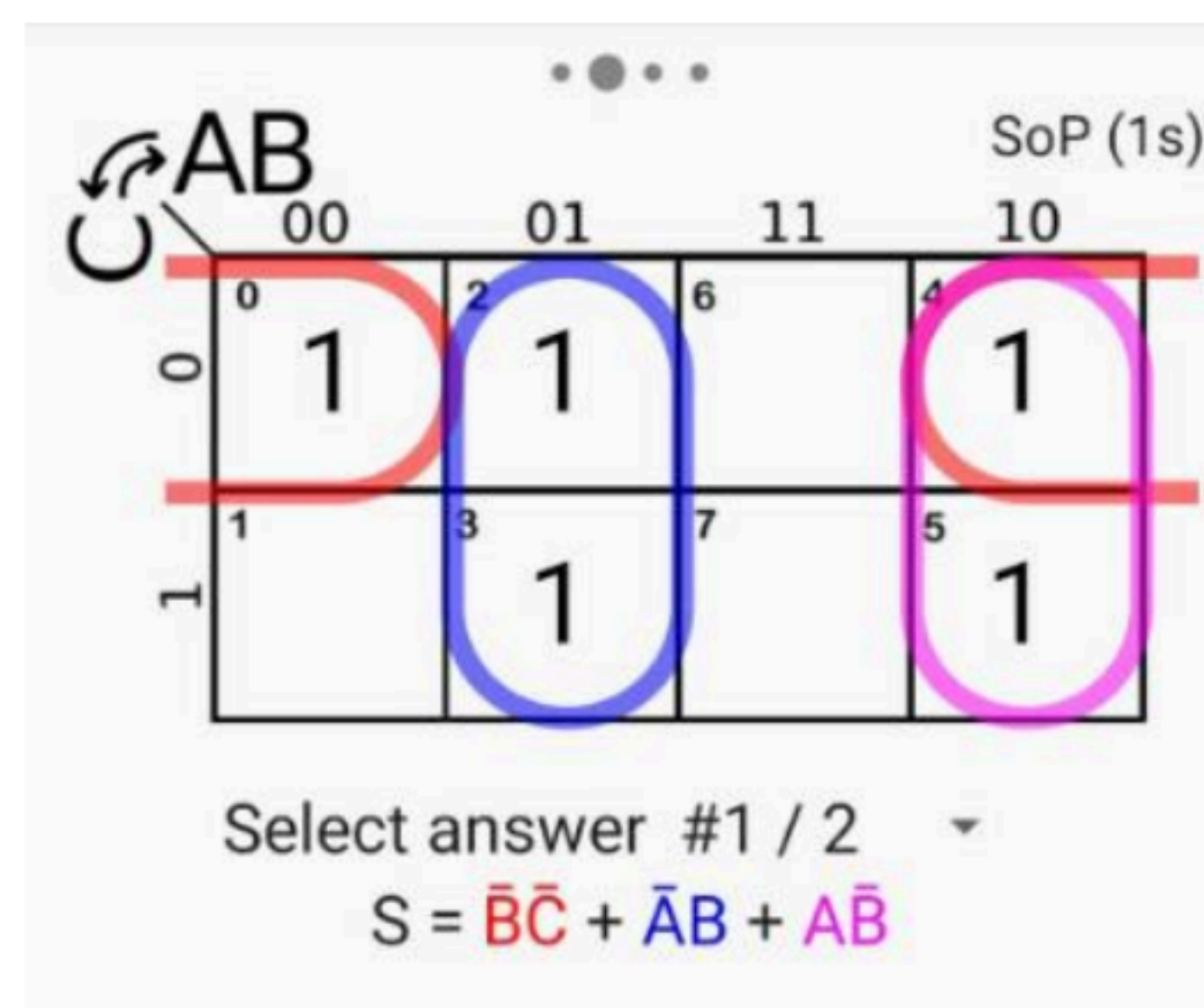
For E



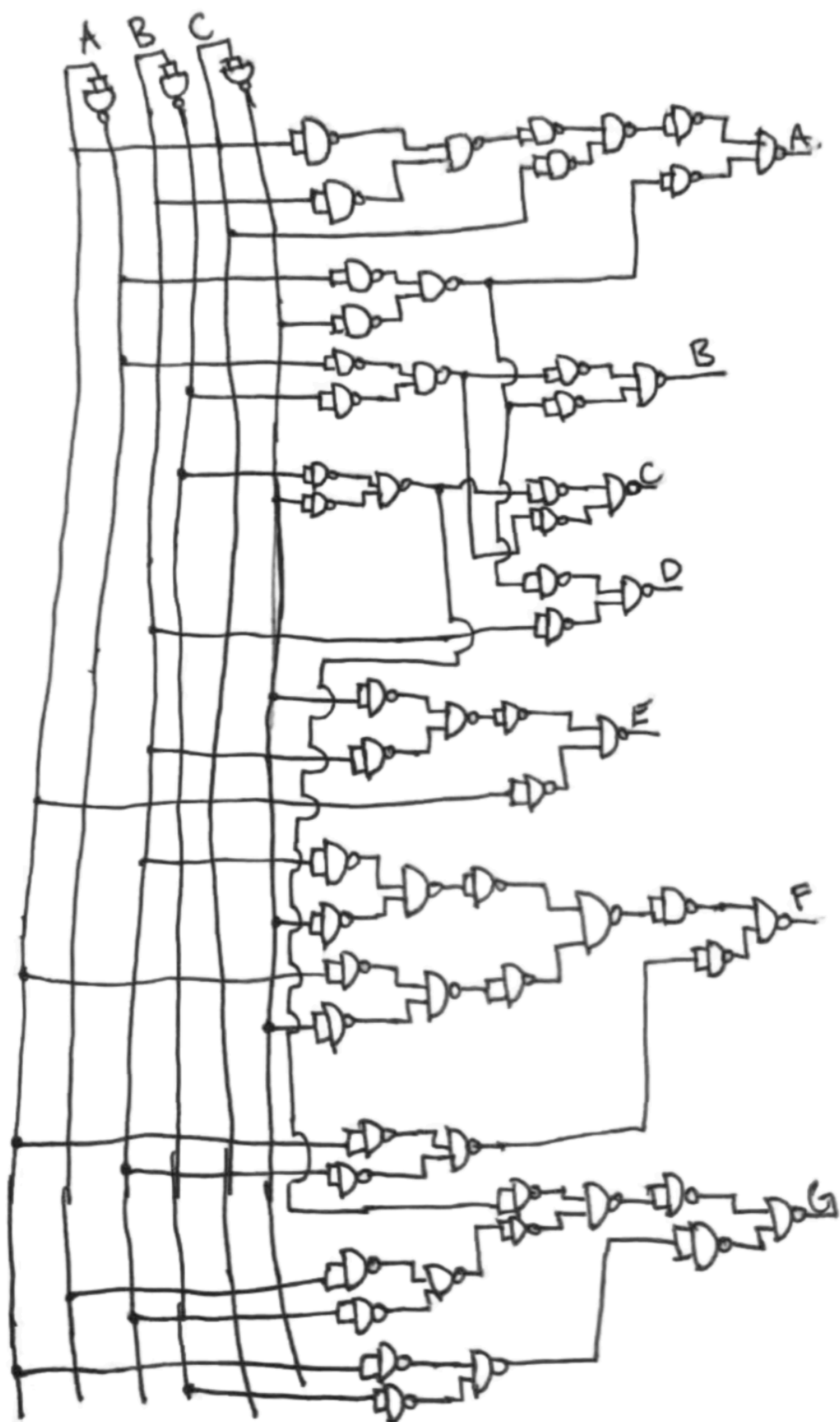
For F



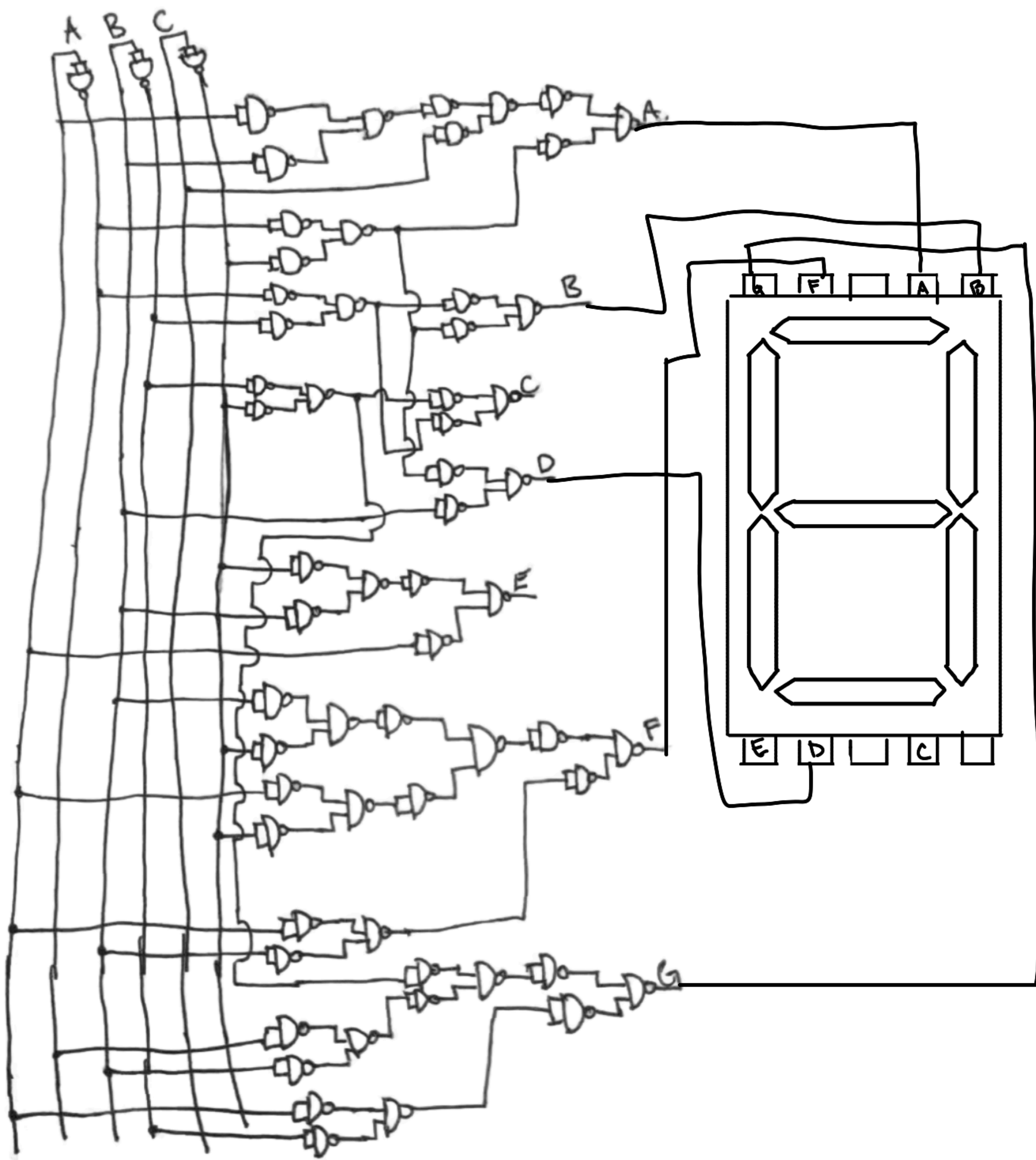
For G



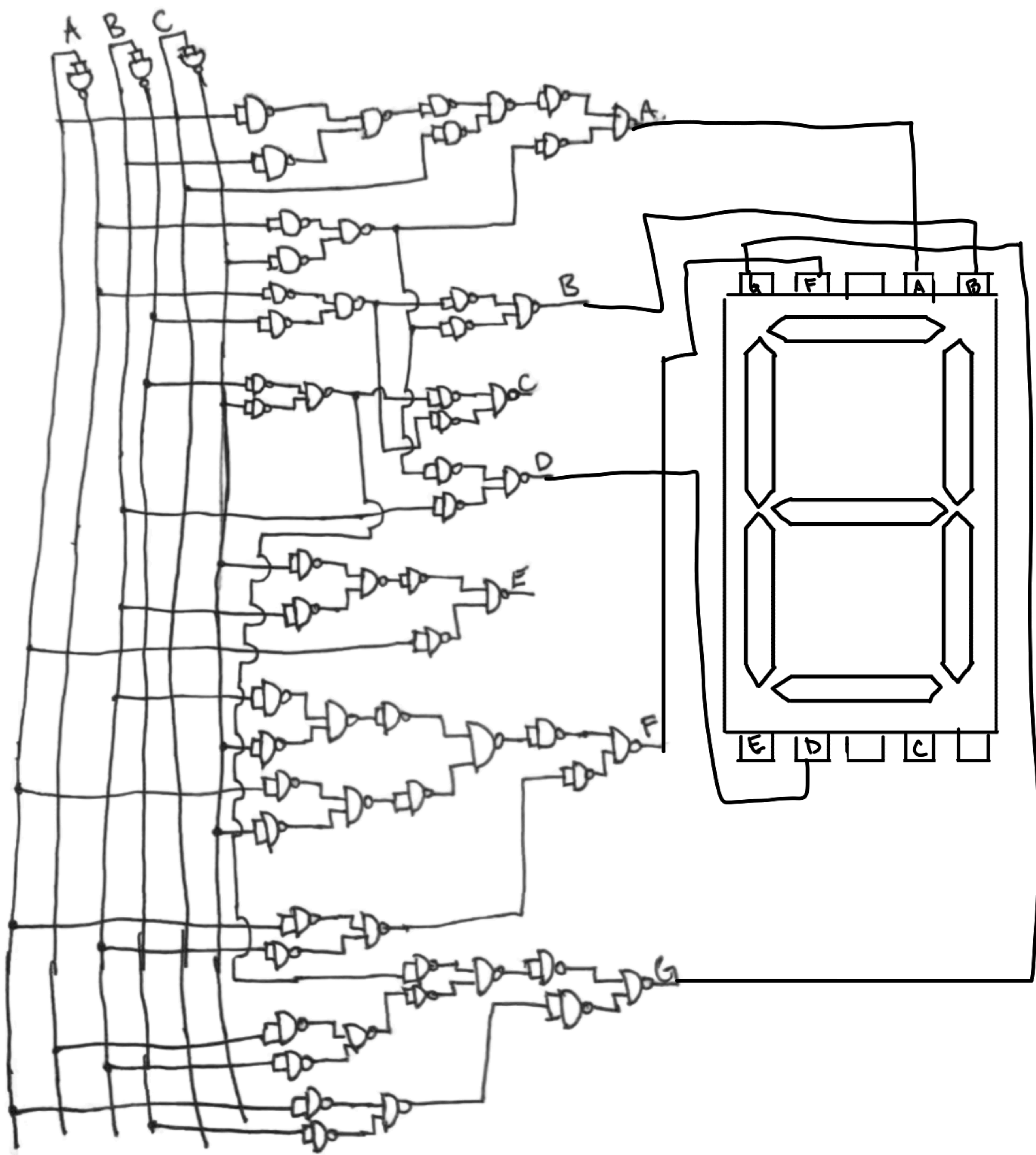
SOP STANDARD /SIMPLIFIED CIRCUIT



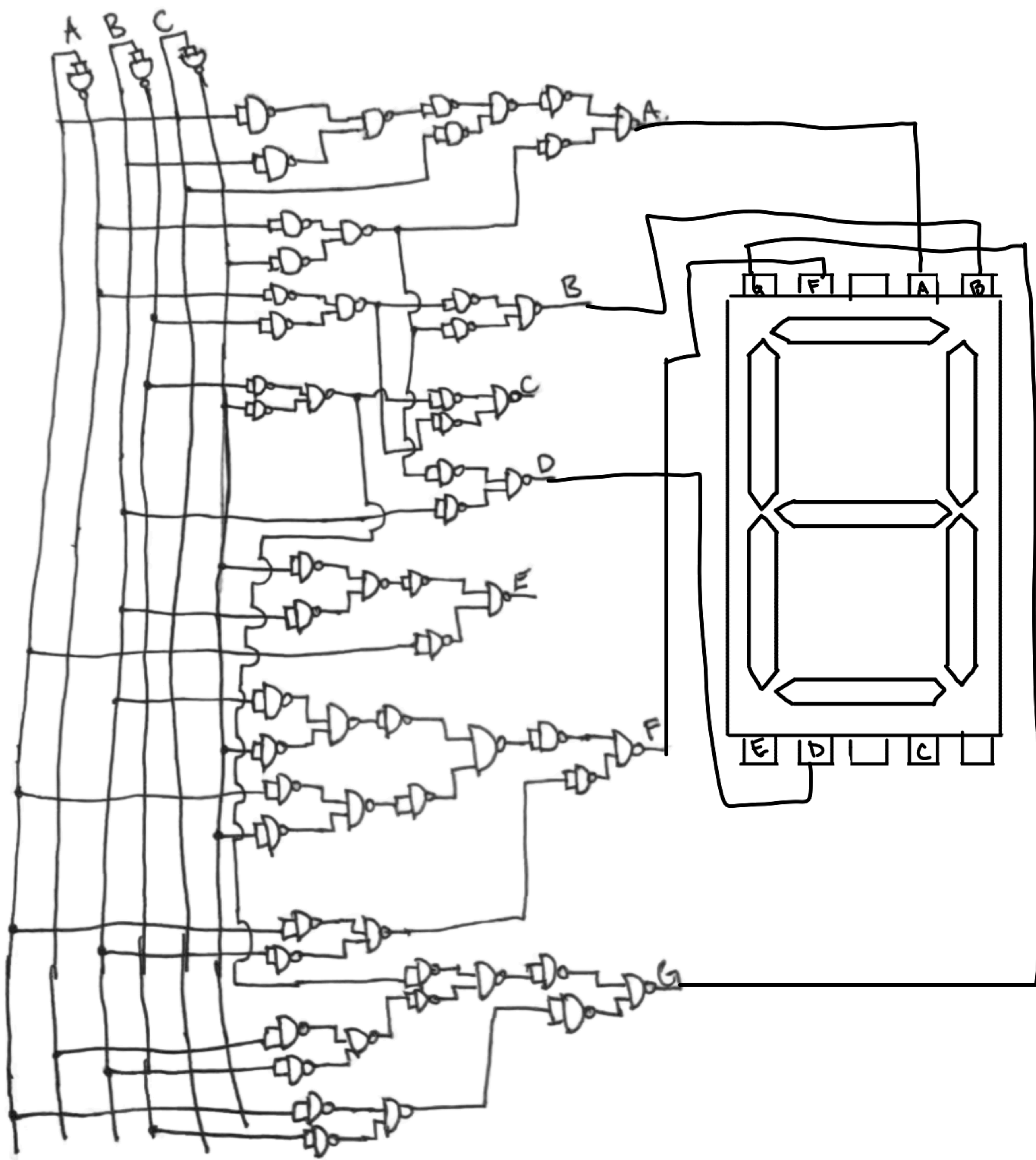
SIMULATION



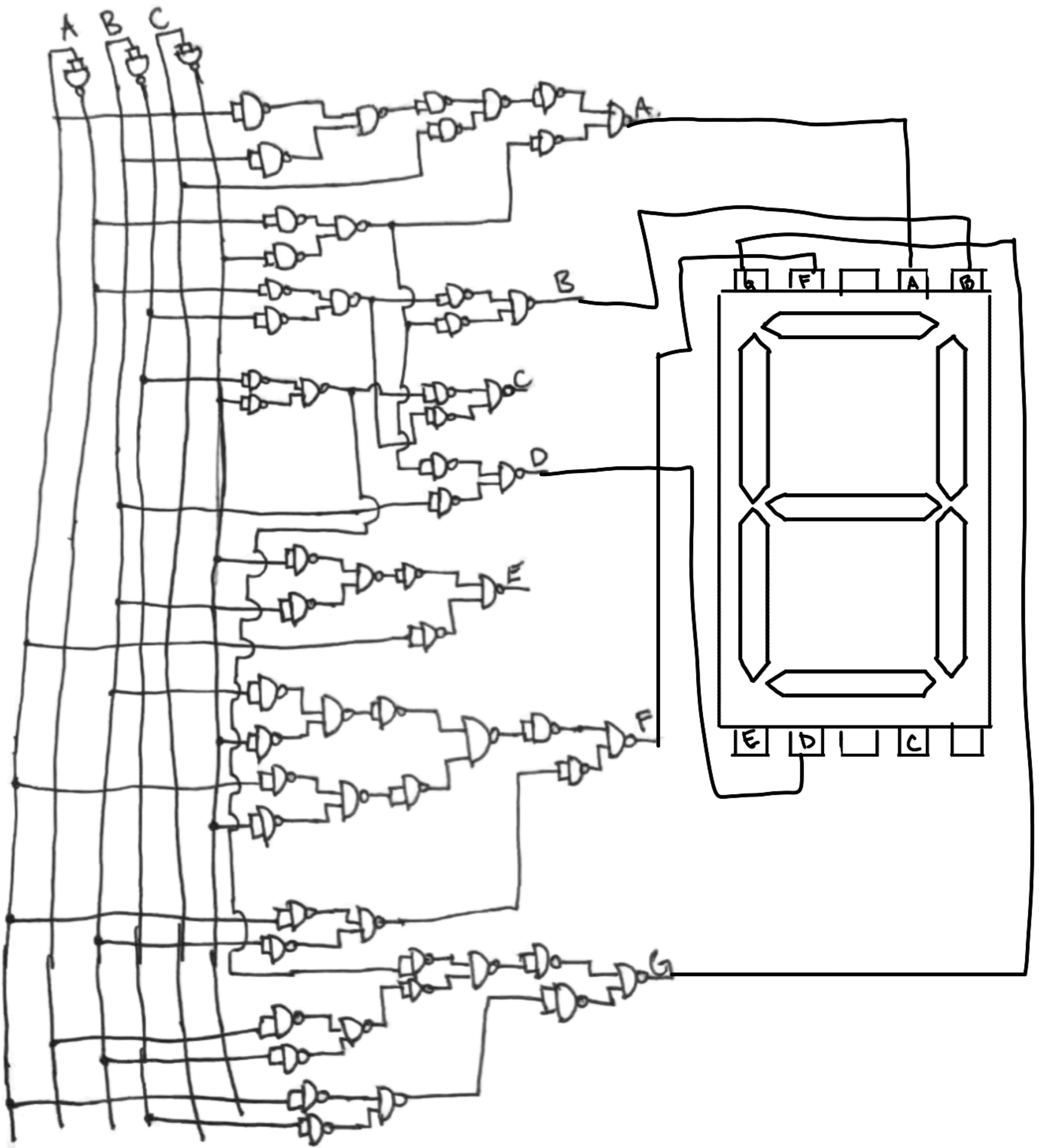
SIMULATION



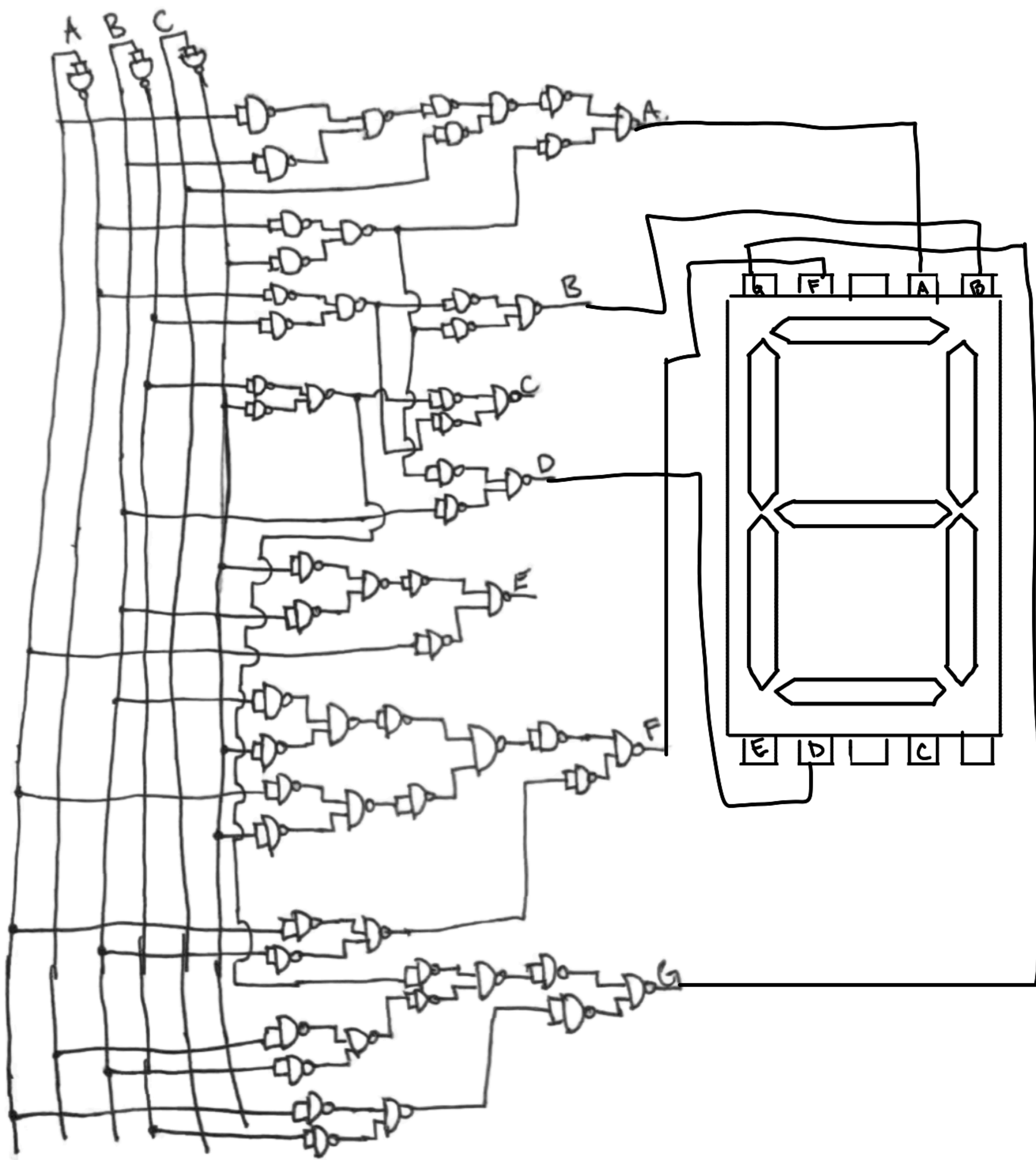
SIMULATION



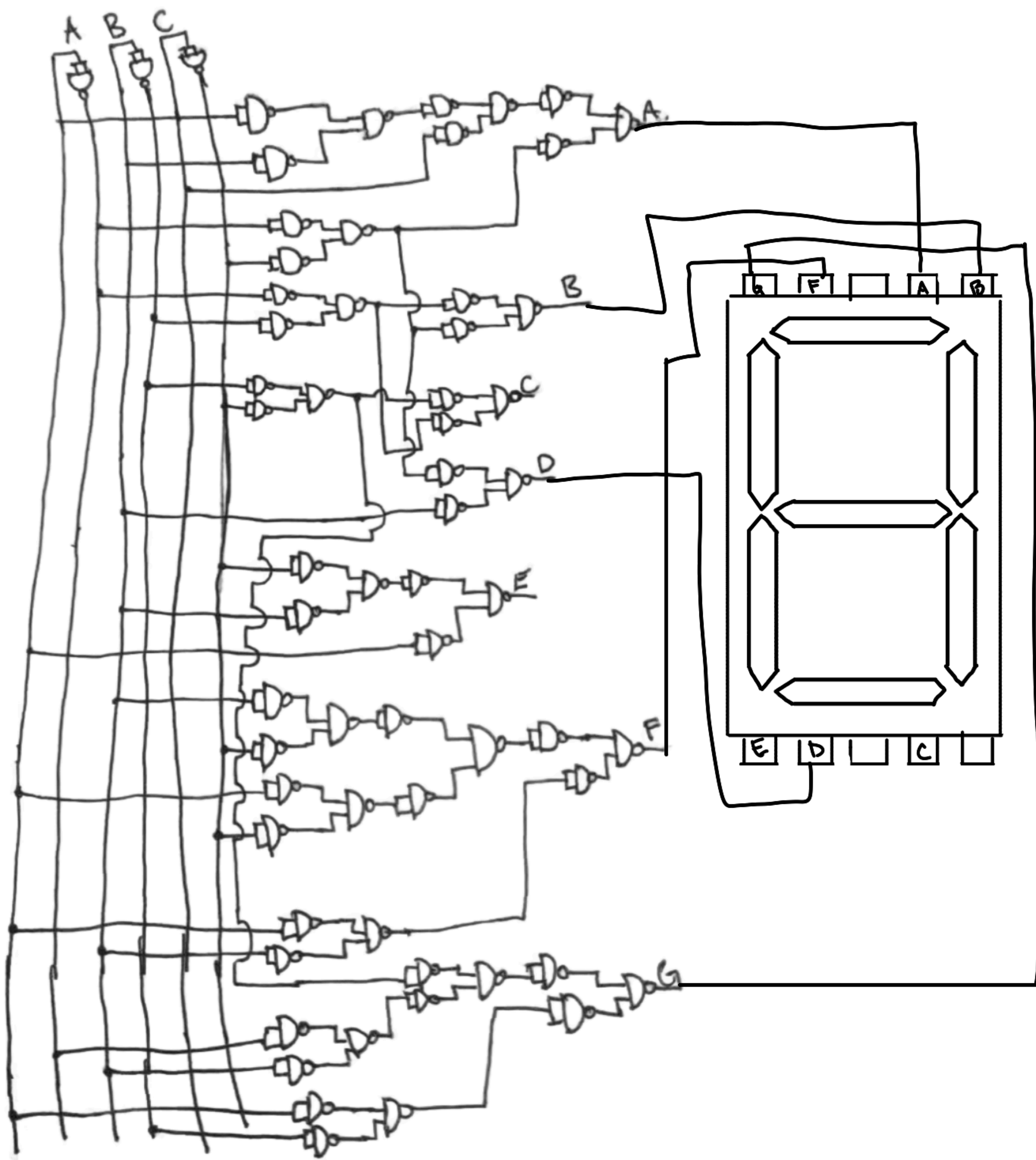
SIMULATION



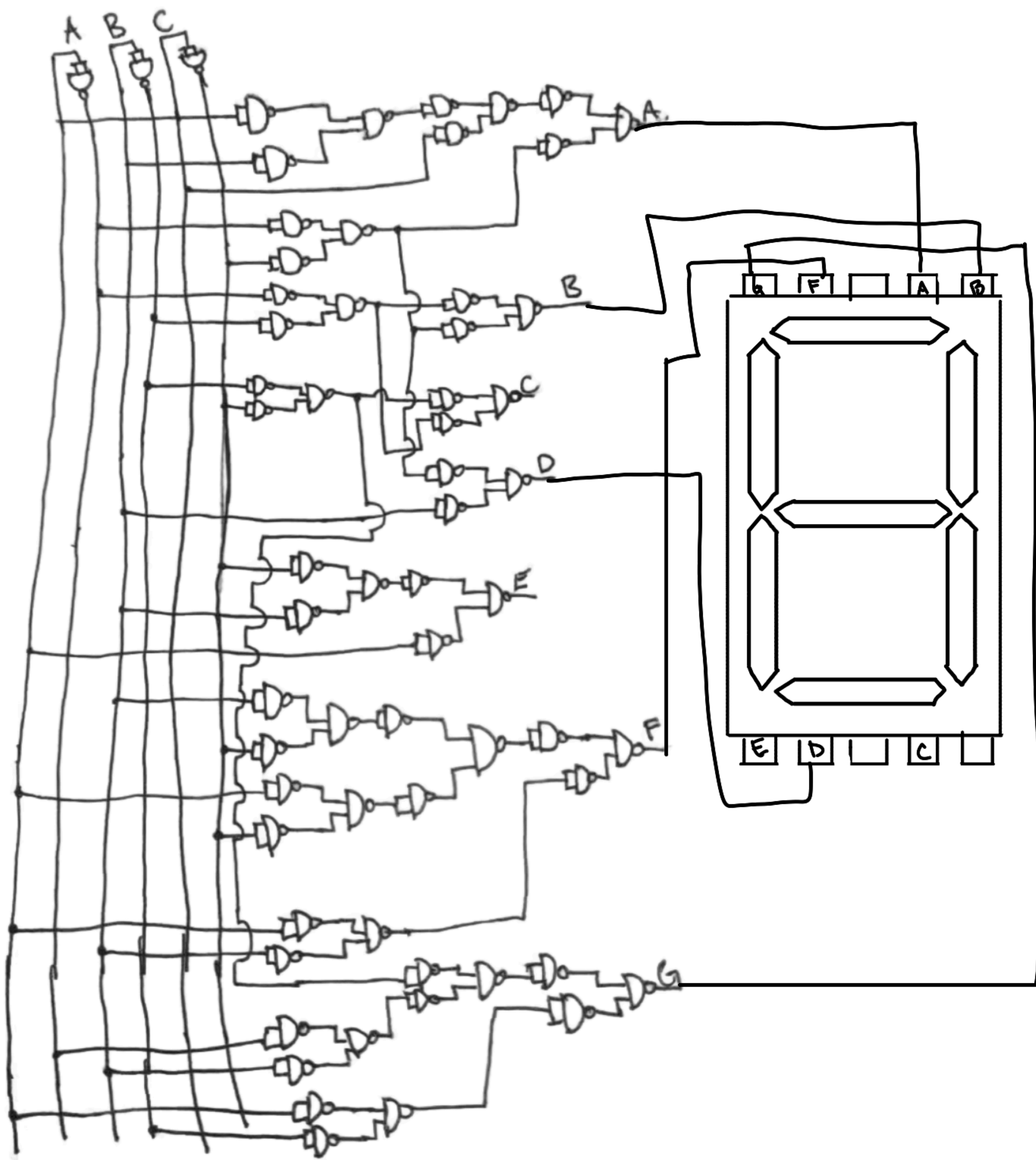
SIMULATION



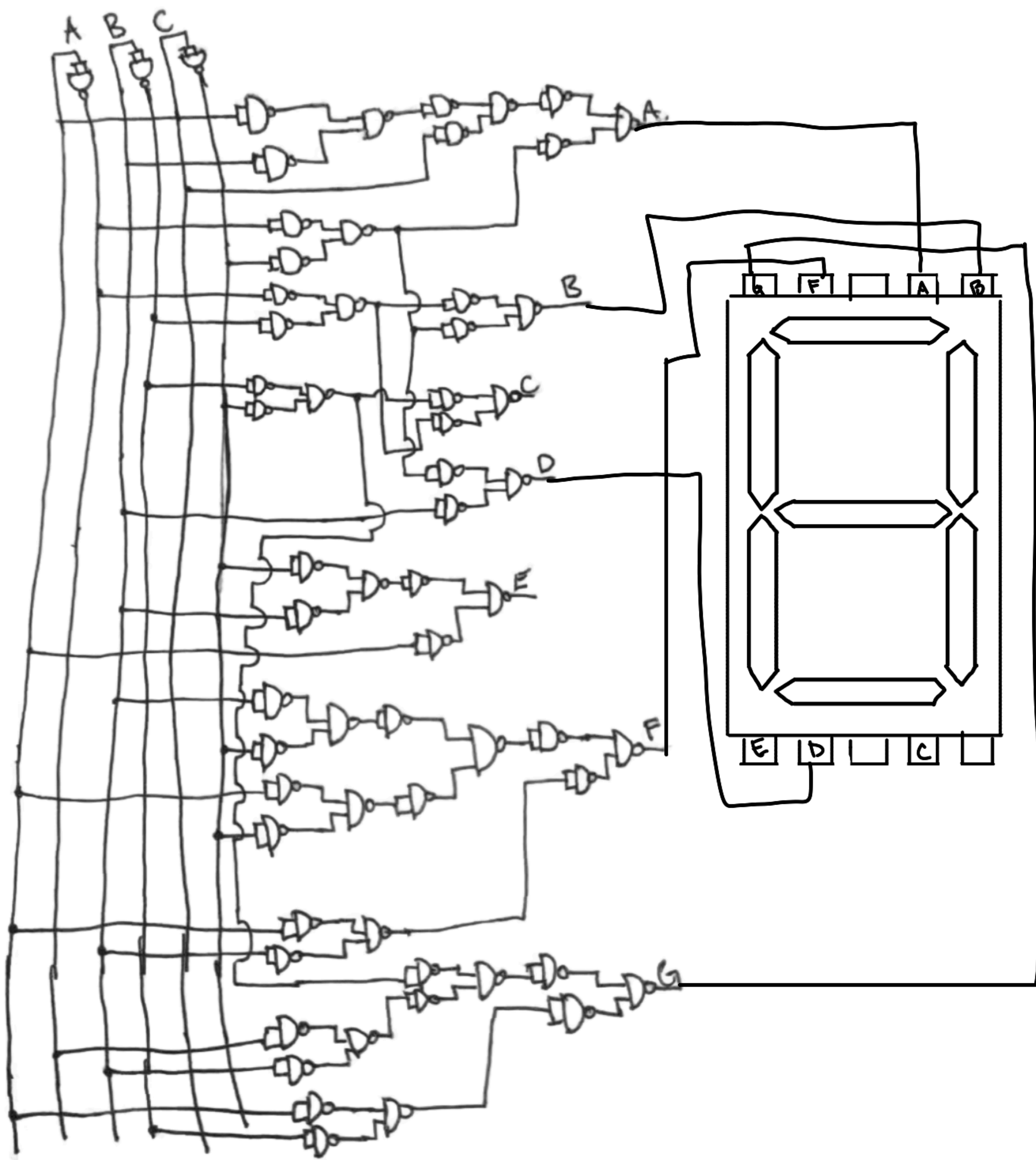
SIMULATION



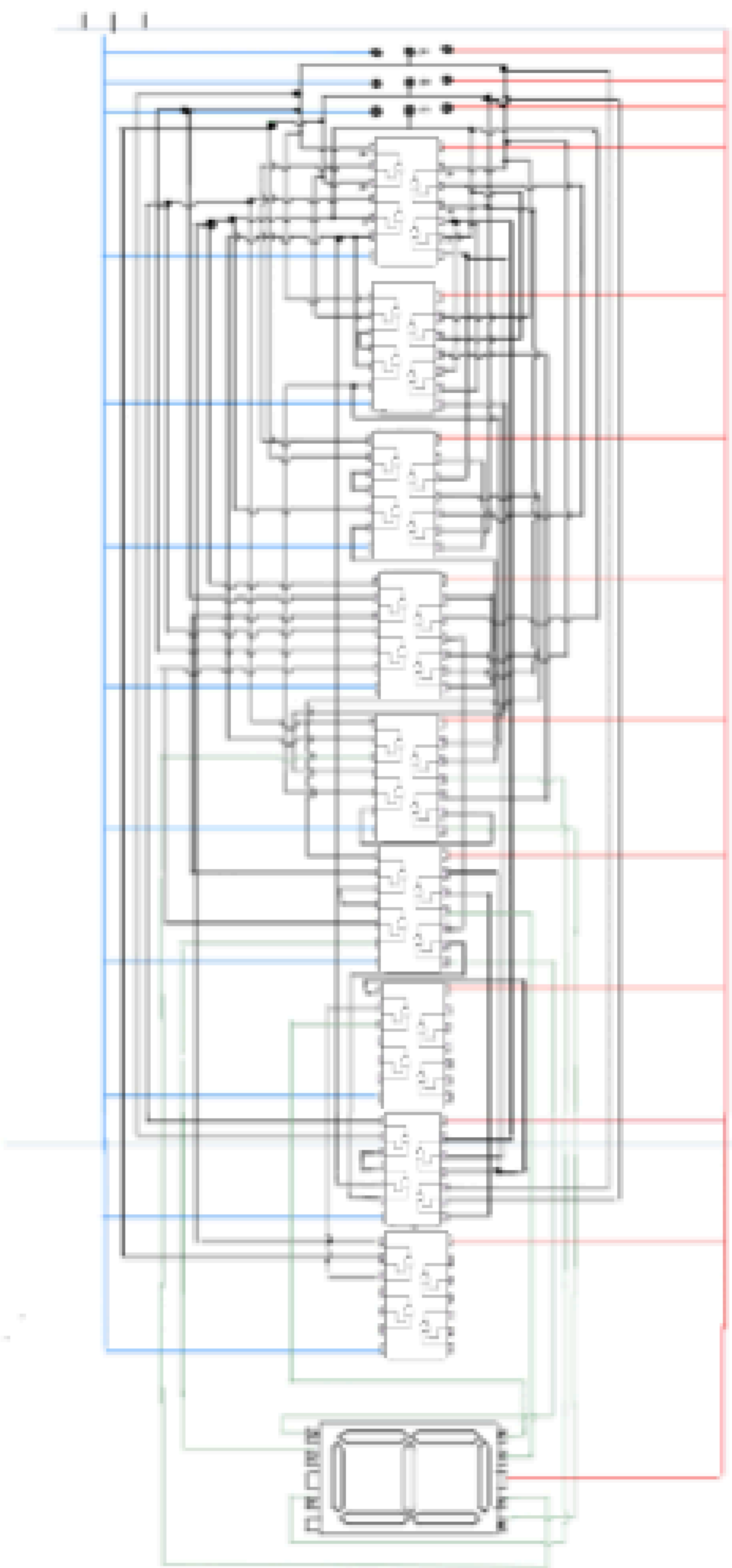
SIMULATION



SIMULATION



BREADBOARD LAYOUT



CONCLUSION

In conclusion, the combinational logic circuit was successfully designed and verified to display the eight-character sequence (**a,l,e,c,h,r,L,C**) driven by a 3-bit binary input. This project successfully demonstrated the practical application of NOR gates as universal logic elements, validating that a multi-output decoder network can be realized entirely without independent AND, OR, or NOT gates. Through strategic character mapping—such as adapting lowercase configurations for 'r' and 'n' to accommodate the limitations of a standard 7-segment display—the visual output remained highly legible and distinct for every state. By leveraging Product-of-Sums optimization and De Morgan's laws, the boolean expressions were cleanly converted into NOR logic expressions, proving the accuracy of the theoretical truth tables and confirming the reliability of the universal gate implementation.